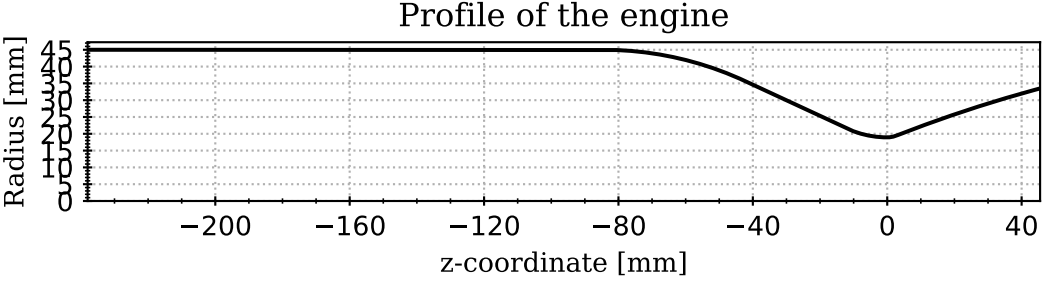
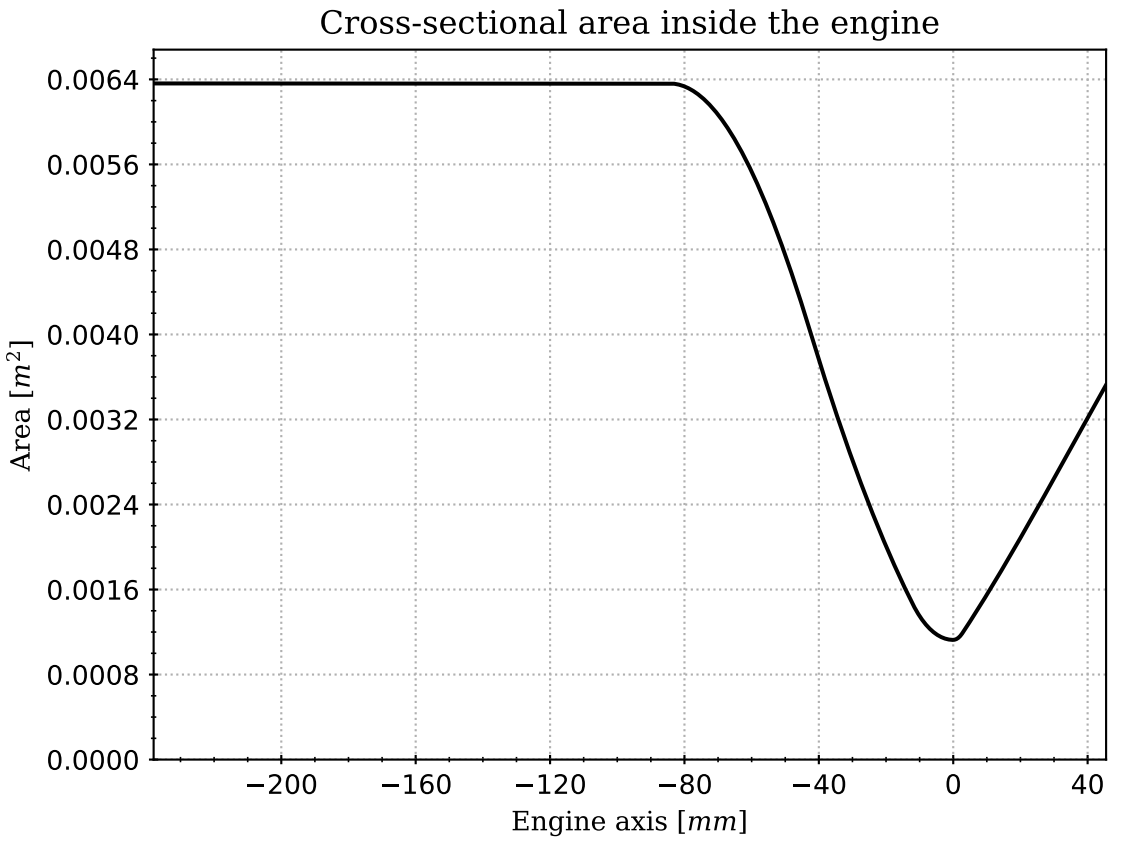
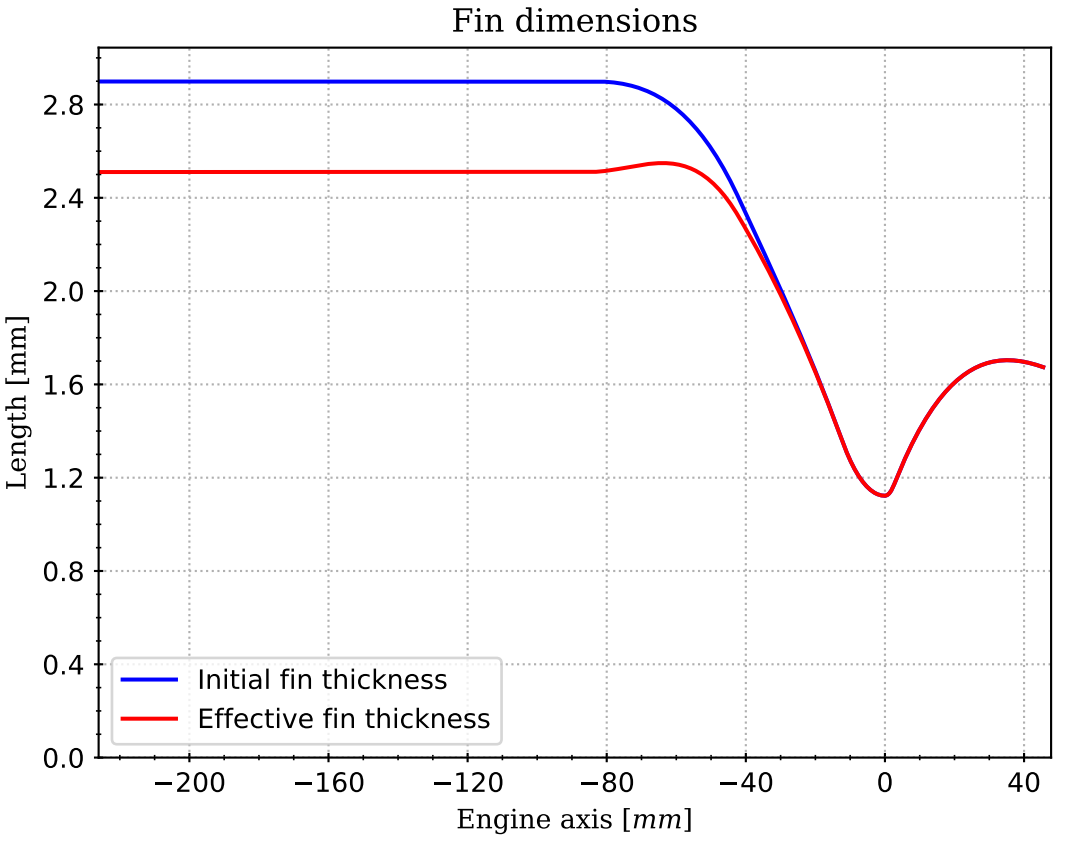
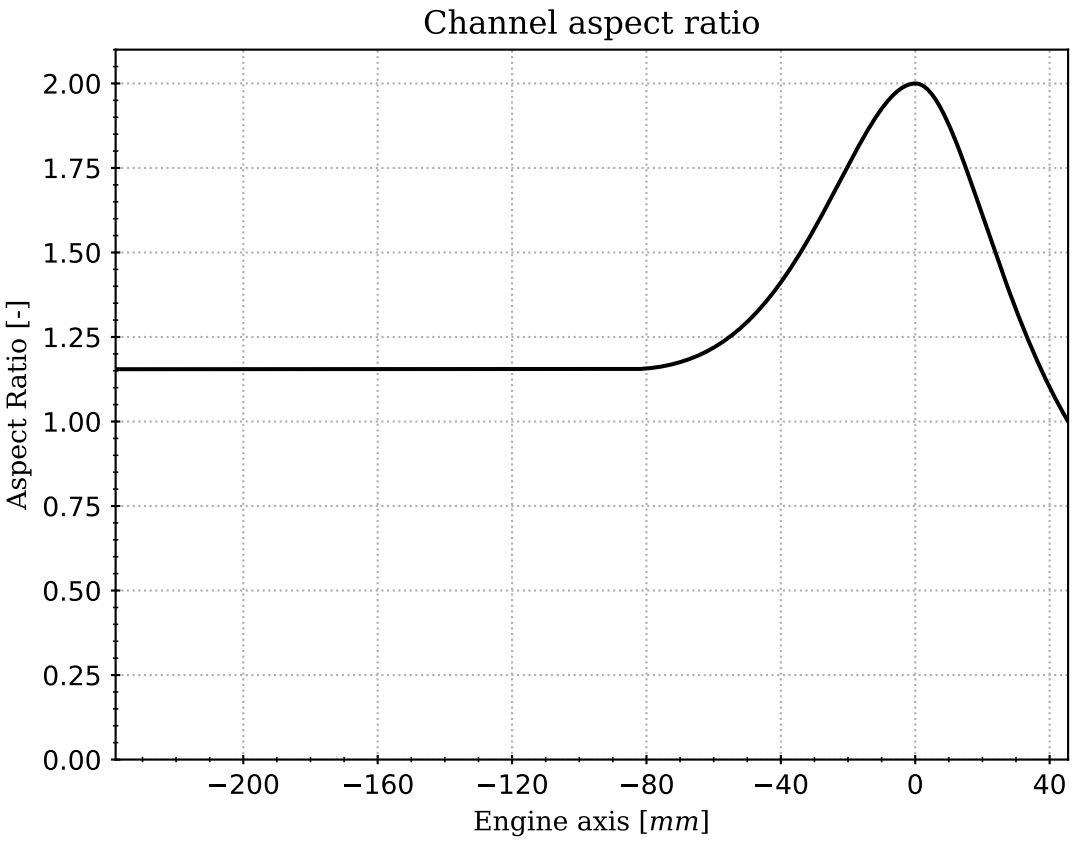
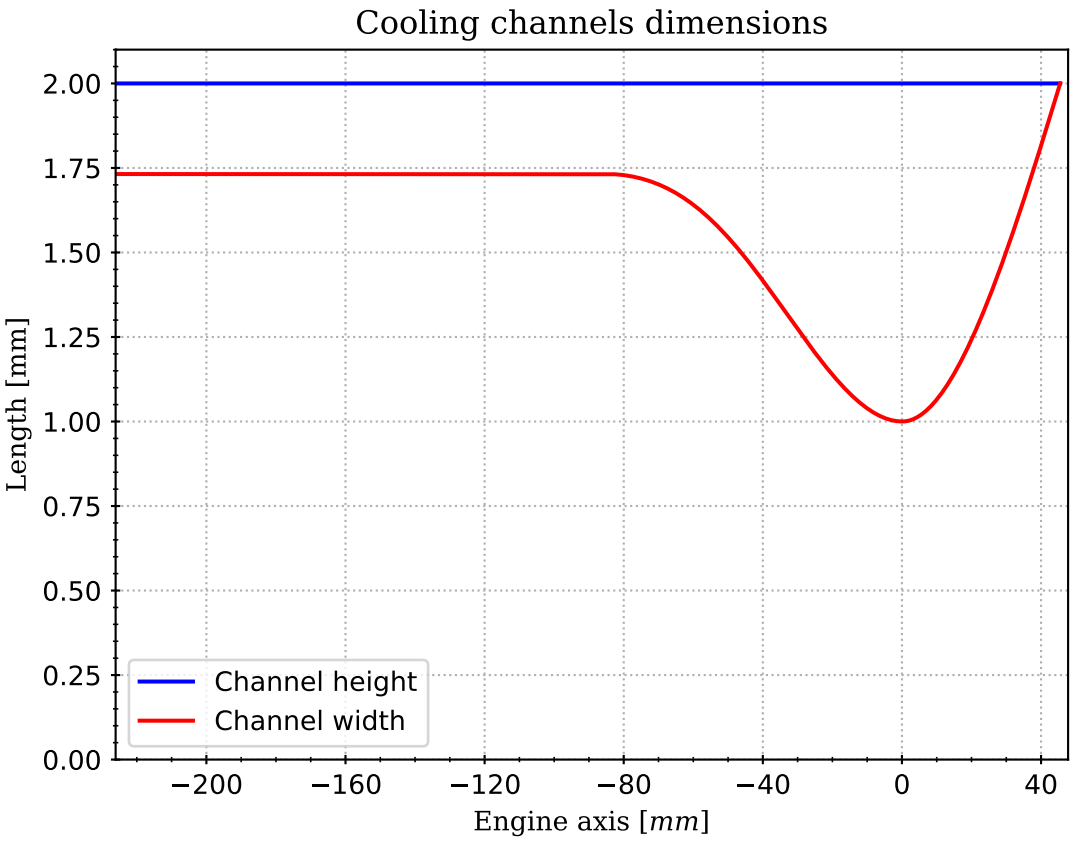
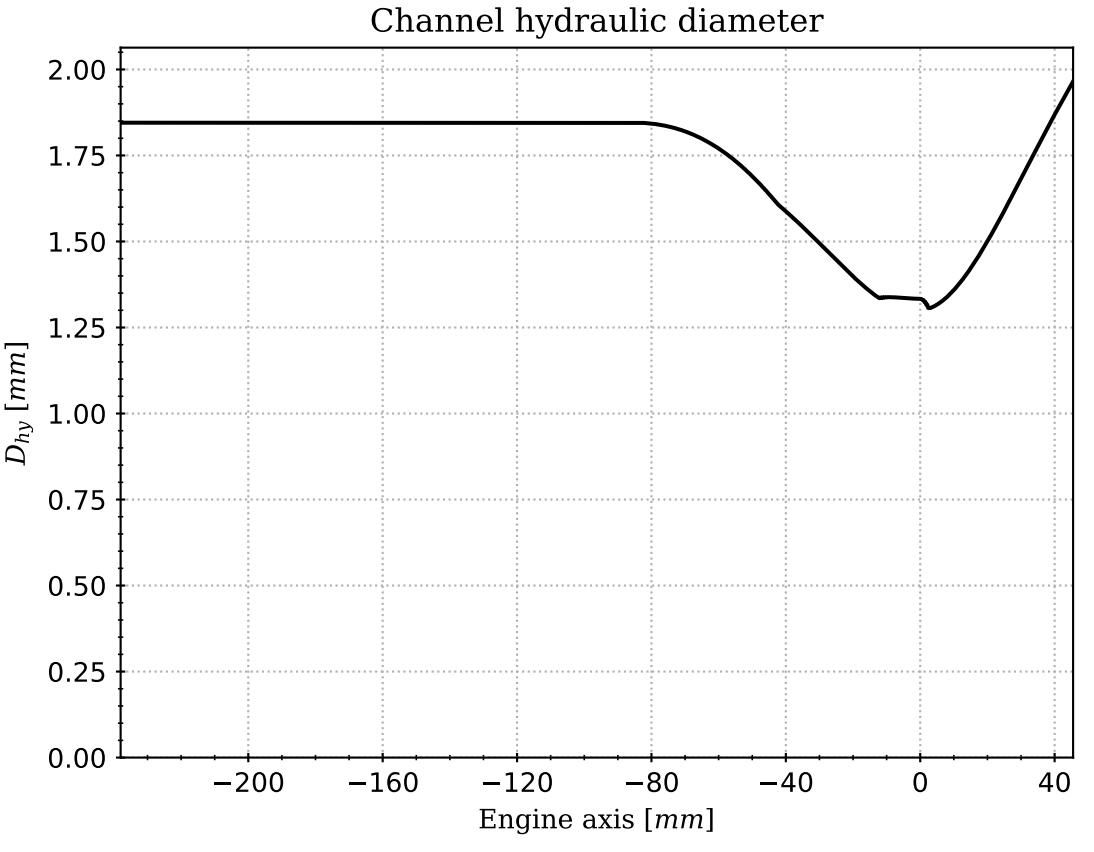

Profile of the engine
Radius [mm]
45
40
35
30
25
20
15
10
5
0
−200
−160
−120
−80
−40
0
40
z-coordinate [mm]

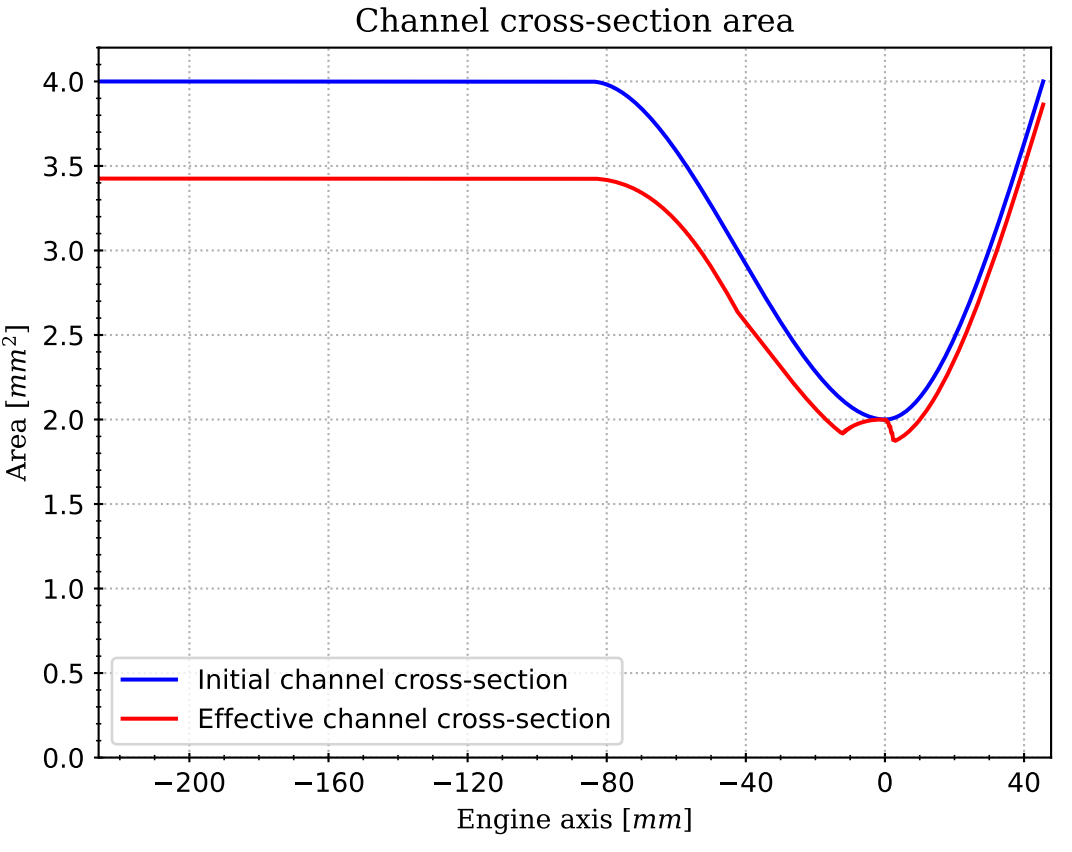

Cross-sectional area inside the engine
0.0064
0.0056
0.0048
0.0040
0.0032
0.0024
0.0016
0.0008
0.0000
Area [m^2]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

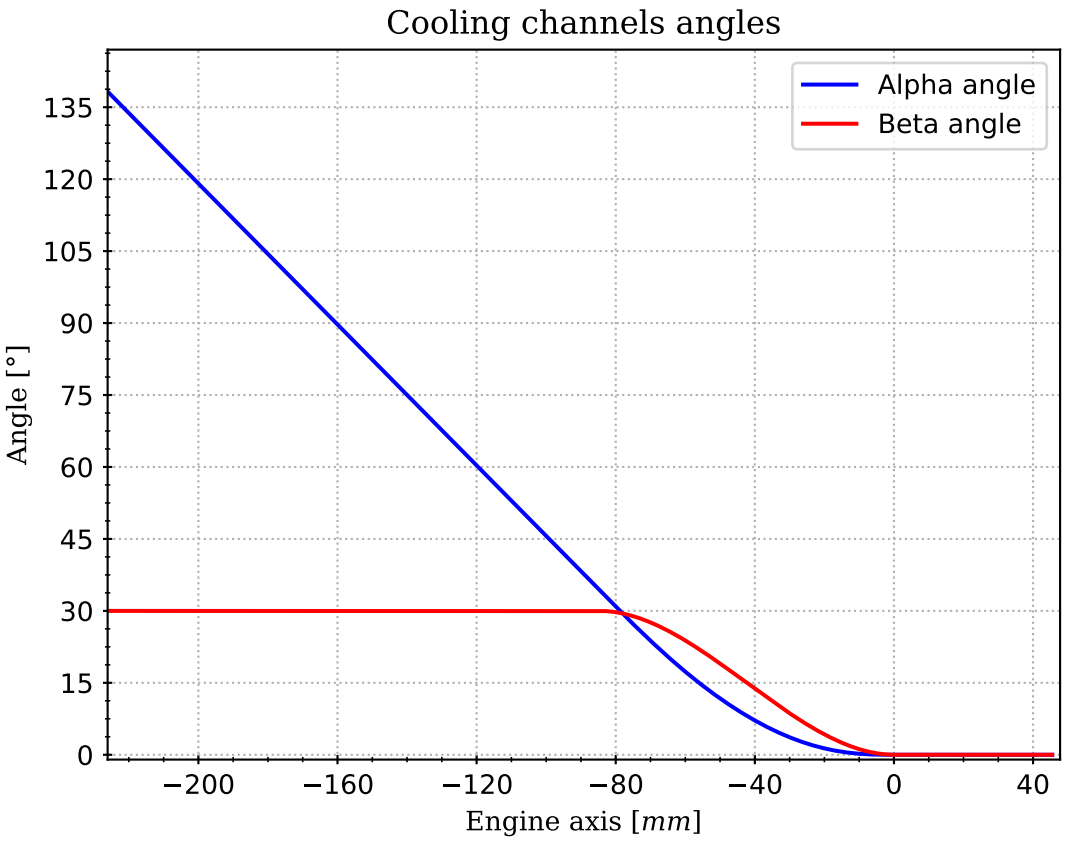

Fin dimensions
Length [mm]
2.8
2.4
2.0
1.6
1.2
0.8
0.4
0.0
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
Initial fin thickness
Effective fin thickness

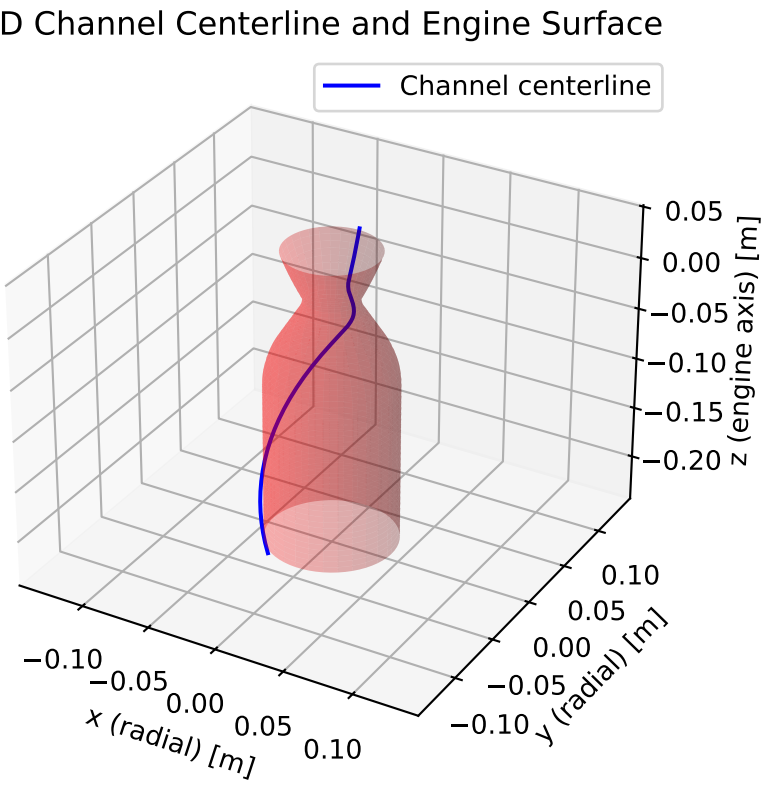

Channel aspect ratio
Aspect Ratio [-]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

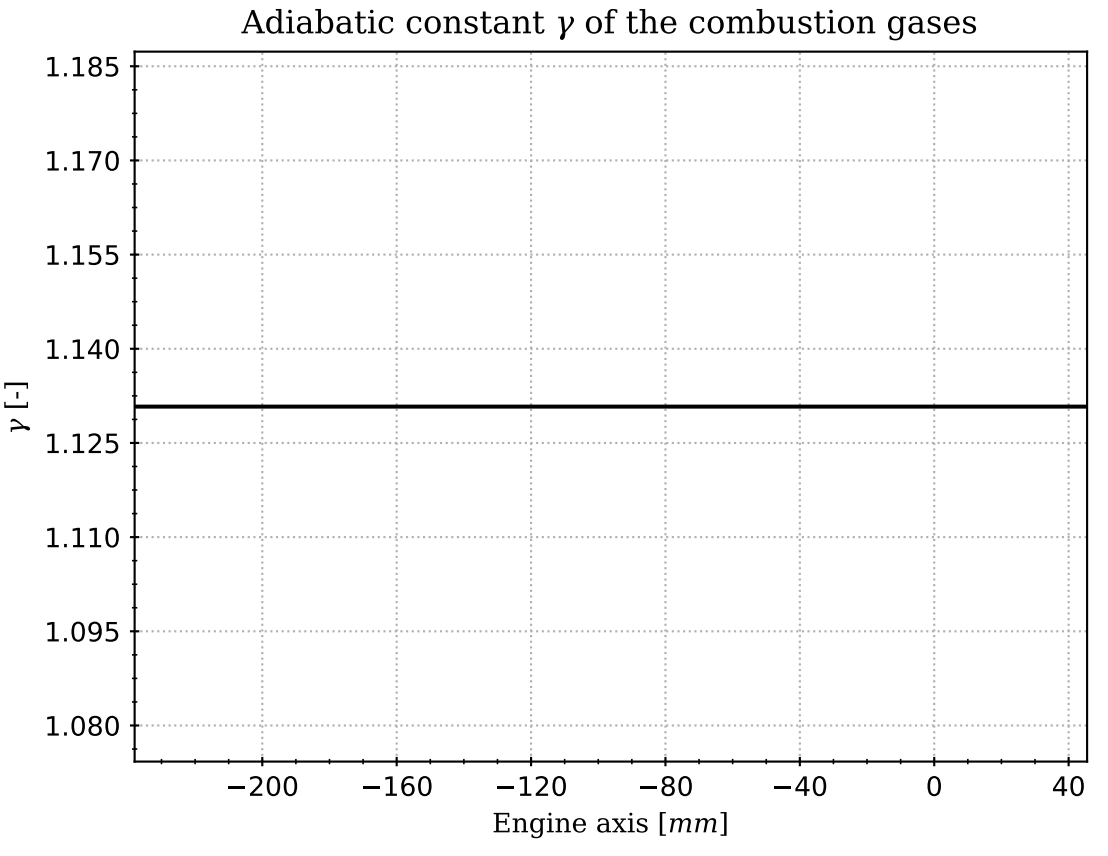

Cooling channels dimensions
Length [mm]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
Channel height
Channel width

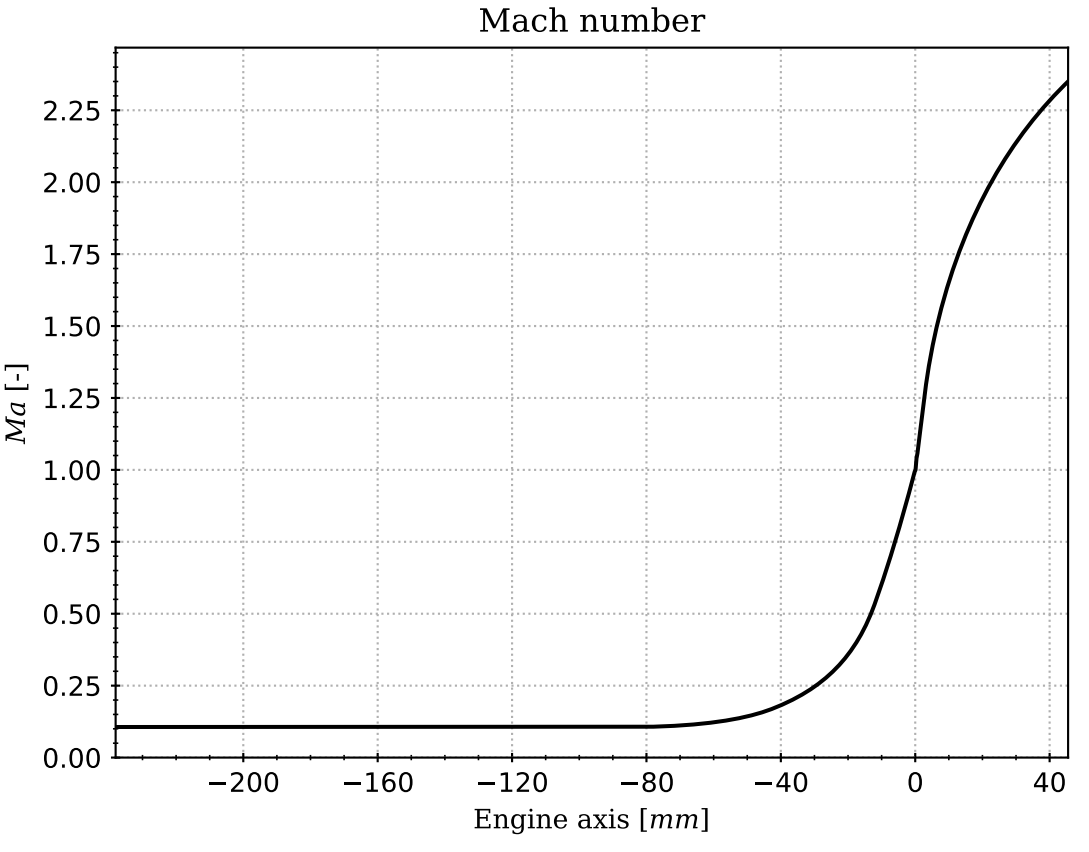

Channel hydraulic diameter
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
D_{hy} [mm]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

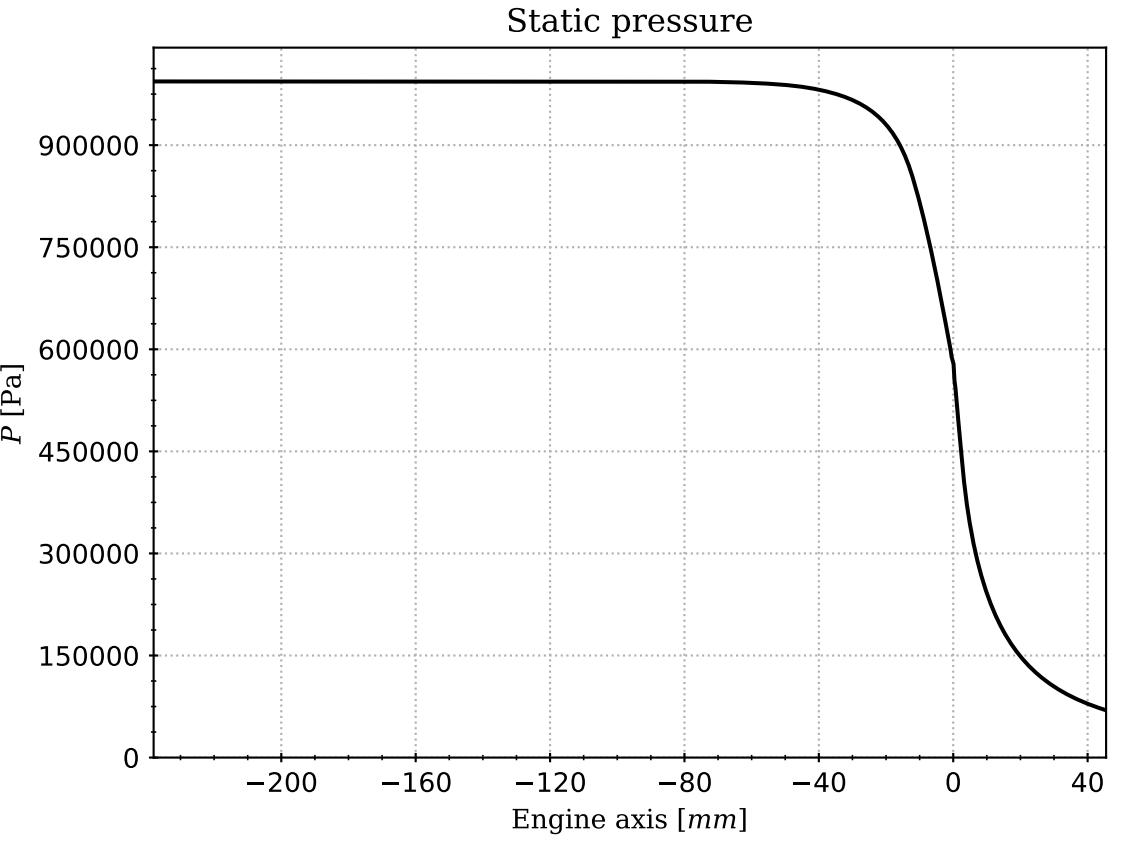

Channel cross-section area
Area [mm^2]
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
Initial channel cross-section
Effective channel cross-section

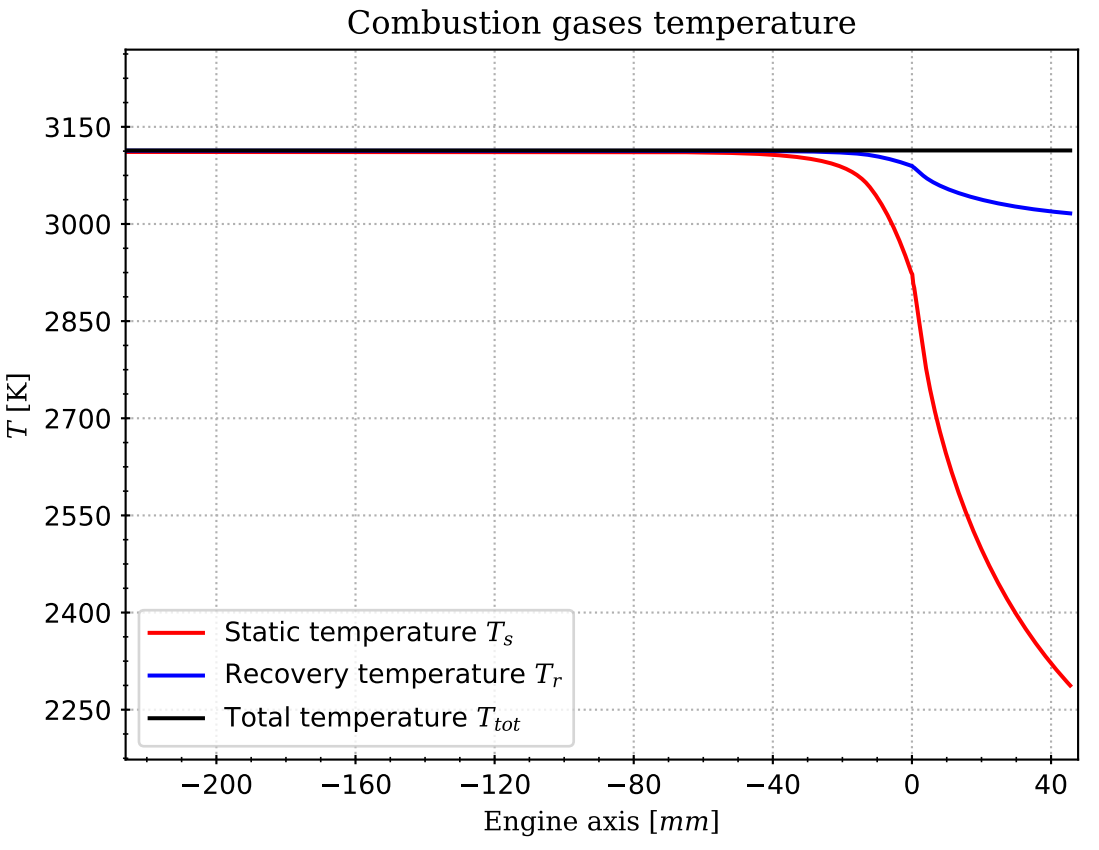

Cooling channels angles
Alpha angle
Beta angle
Angle [°]
135
120
105
90
75
60
45
30
15
0
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

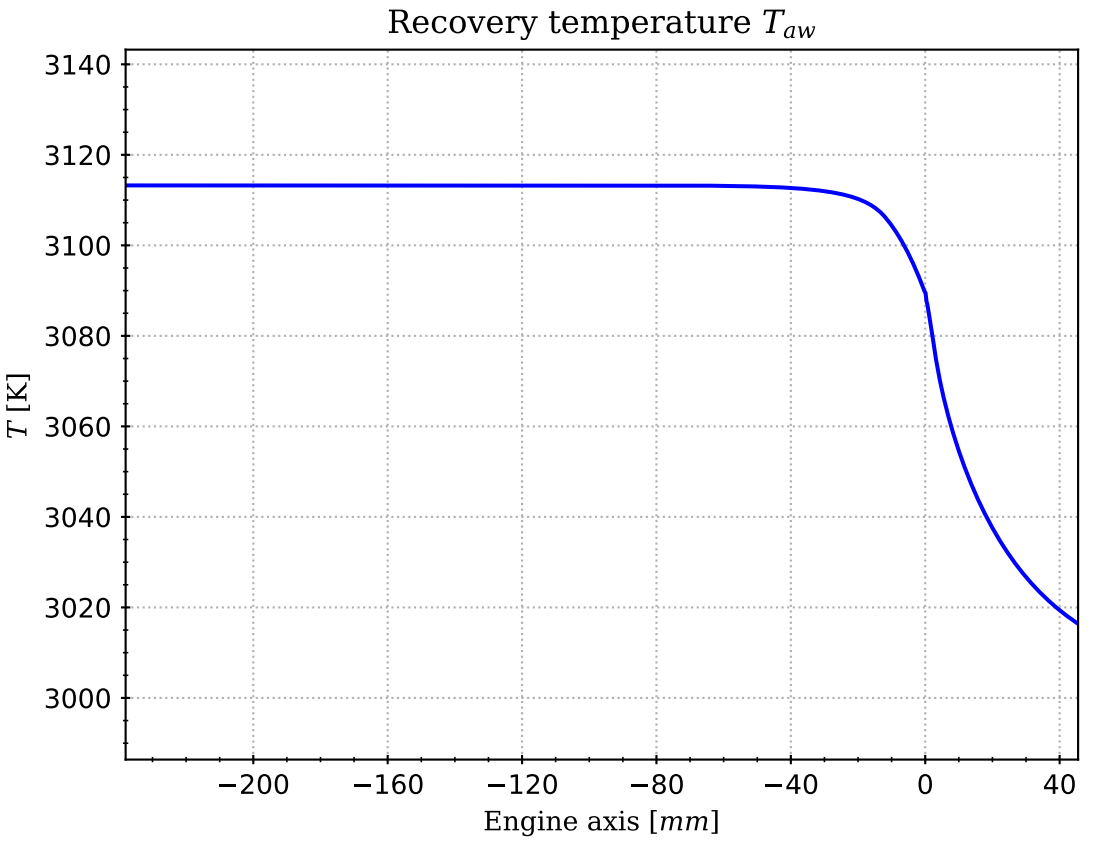

3D Channel Centerline and Engine Surface
Channel centerline
0.05
0.00
−0.05
−0.10
−0.15
−0.20
z (engine axis) [m]
0.10
0.05
0.00
−0.05
−0.10
y (radial) [m]
−0.10
−0.05
0.00
0.05
0.10
x (radial) [m]

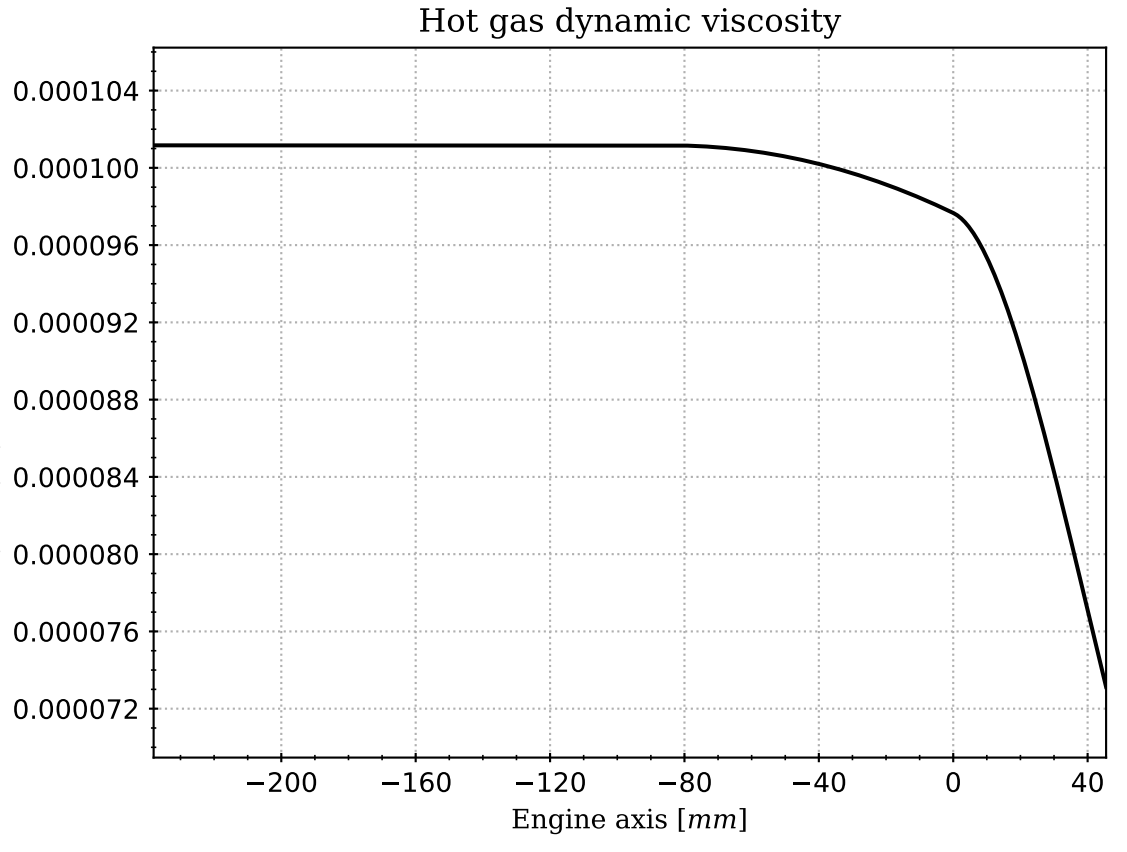

Adiabatic constant γ of the combustion gases
1.185
1.170
1.155
1.140
1.125
1.110
1.095
1.080
γ [-]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

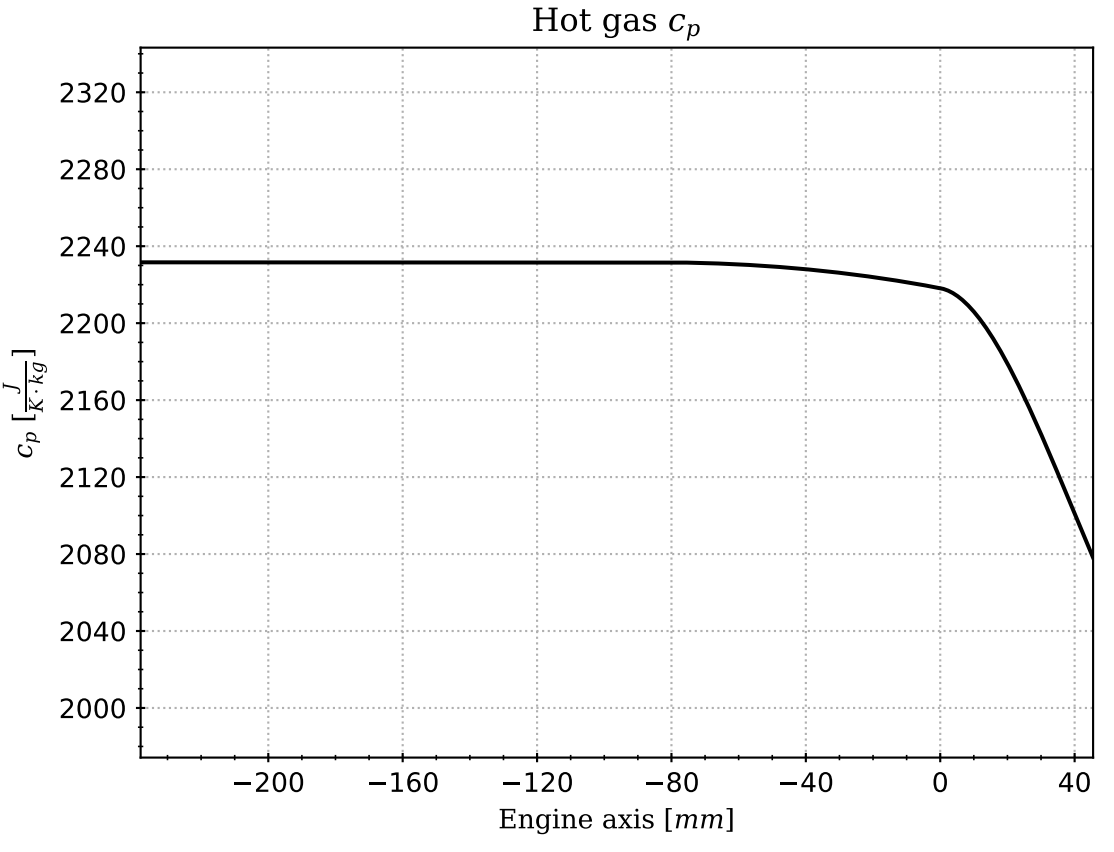

Mach number
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Ma [-]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

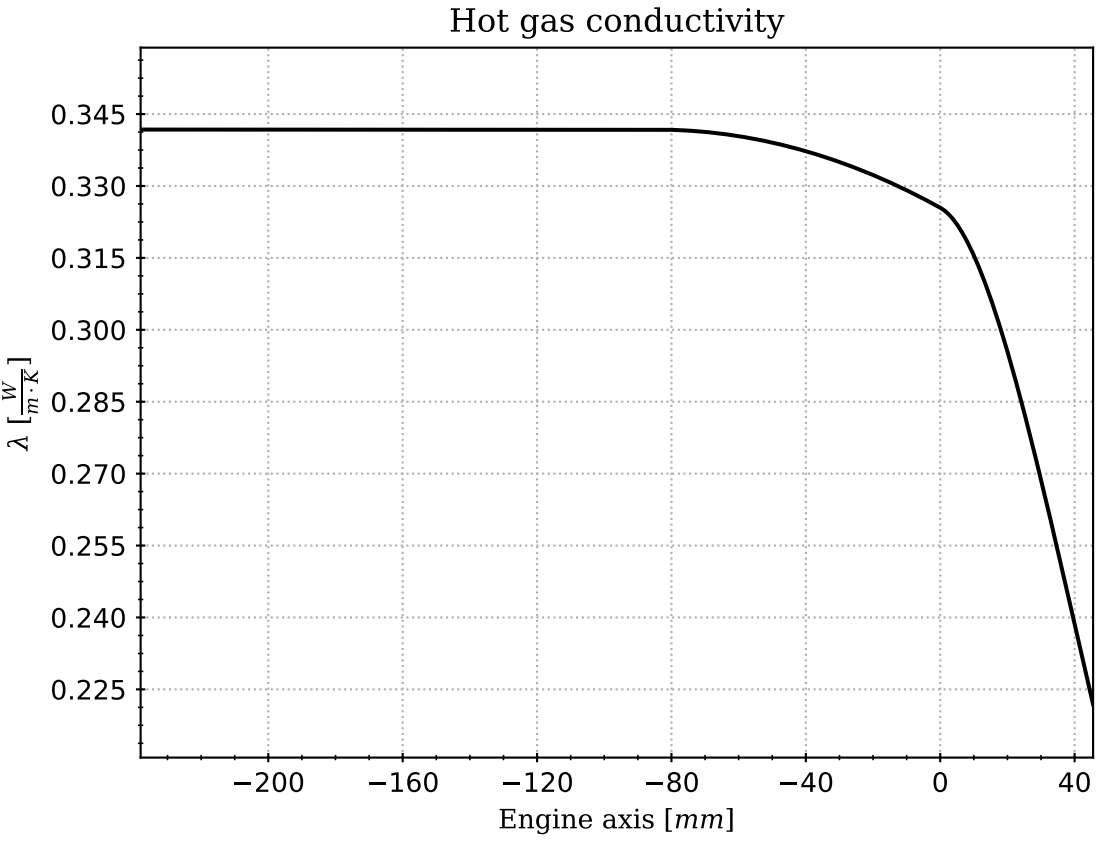

Static pressure
900000
750000
600000
450000
300000
150000
0
P [Pa]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

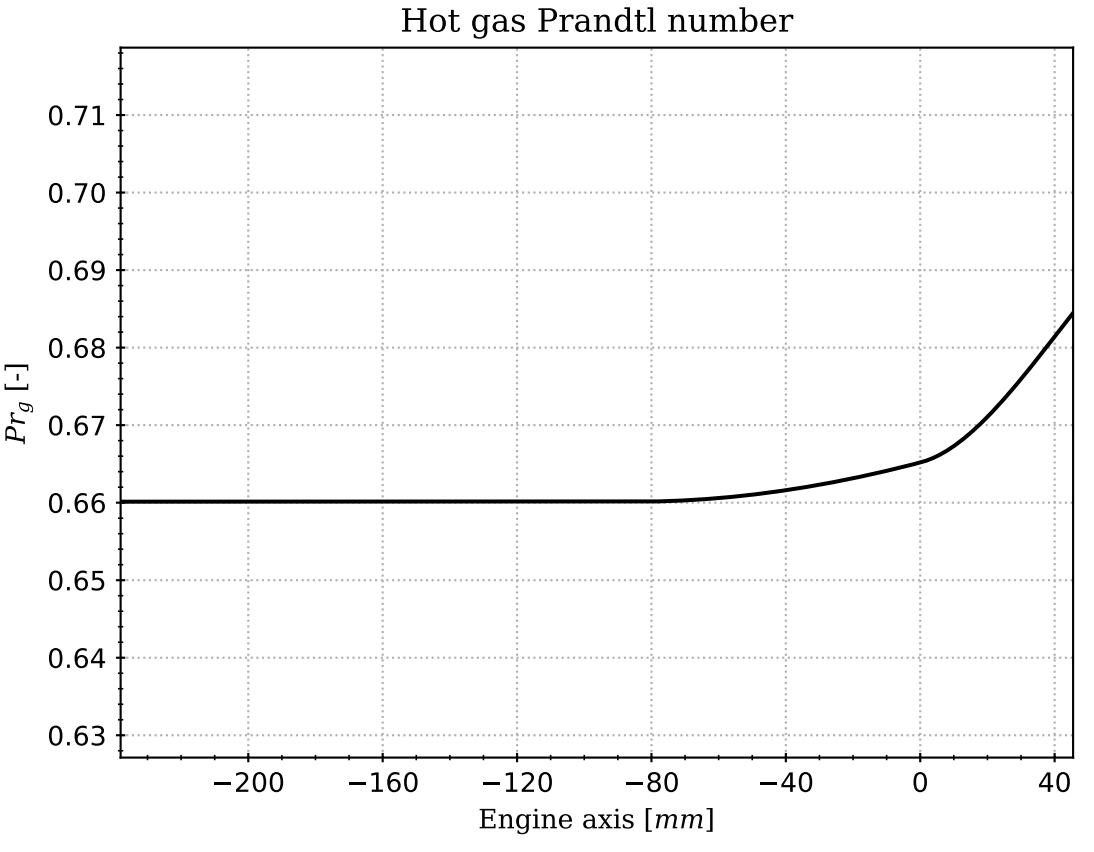

Combustion gases temperature
3150
3000
2850
2700
2550
2400
2250
T [K]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
Static temperature T_s
Recovery temperature T_r
Total temperature T_{tot}

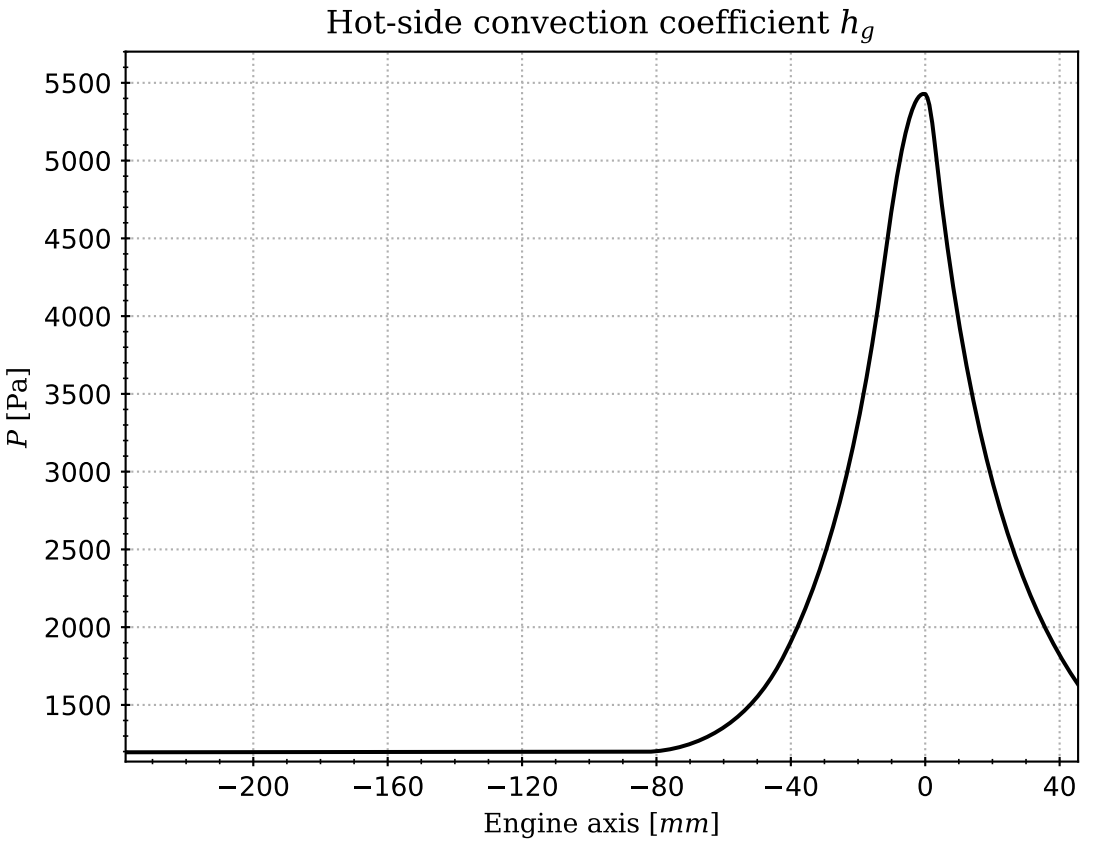

Recovery temperature T_{aw}
3140
3120
3100
3080
3060
3040
3020
3000
T [K]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

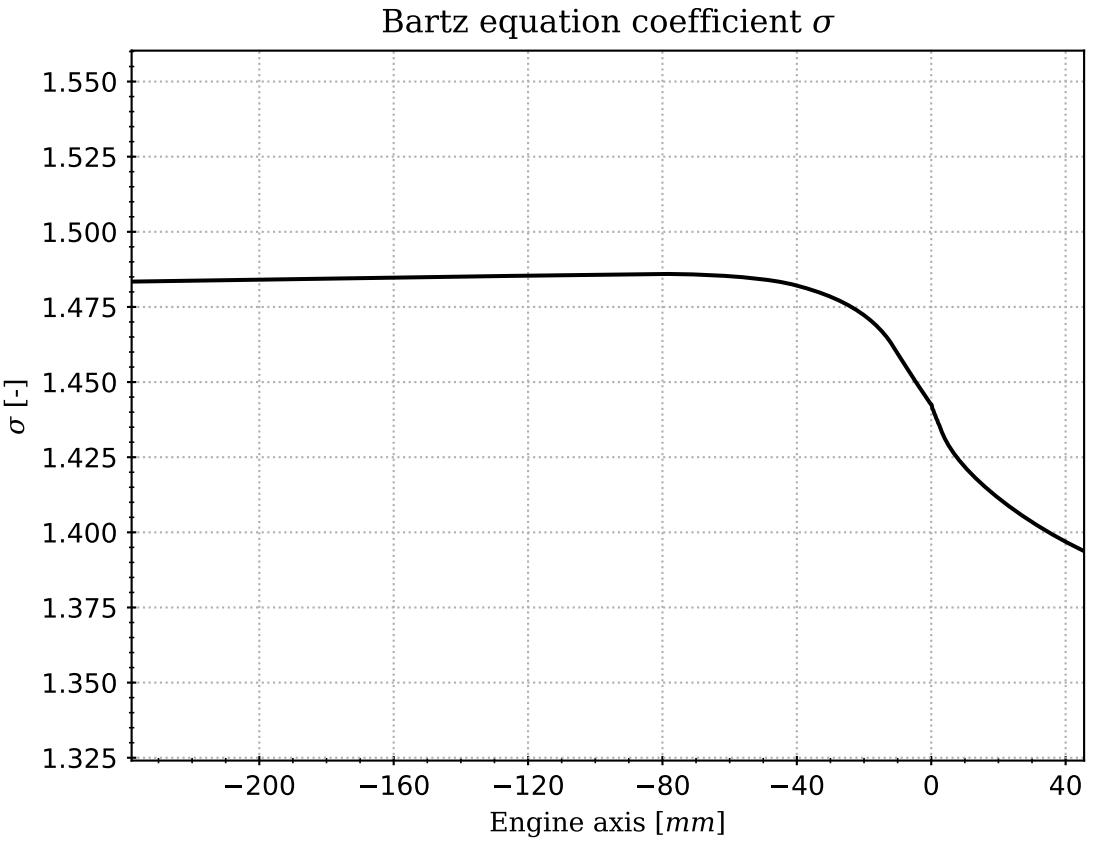

Hot gas dynamic viscosity
0.000104
0.000100
0.000096
0.000092
0.000088
0.000084
0.000080
0.000076
0.000072
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

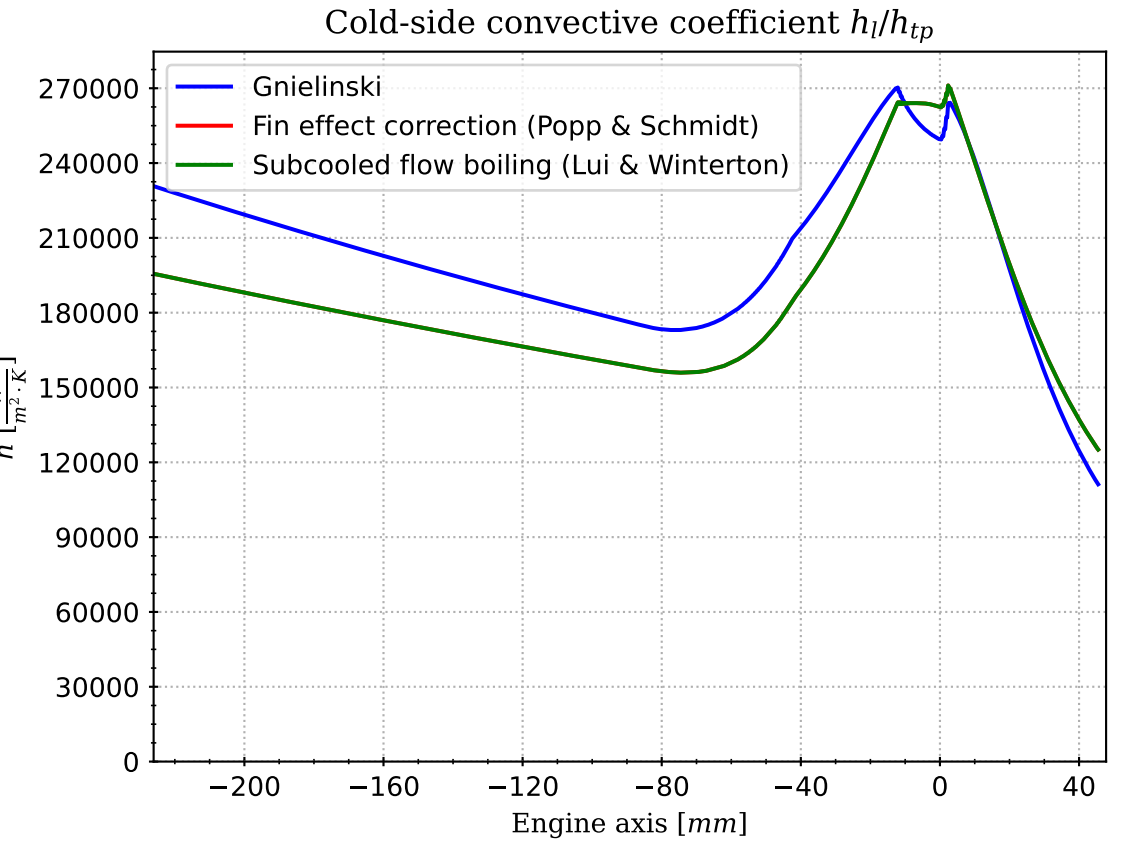

Hot gas c_p
2320
2280
2240
2200
2160
2120
2080
2040
2000
c_p $[\frac{J}{K \cdot kg}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

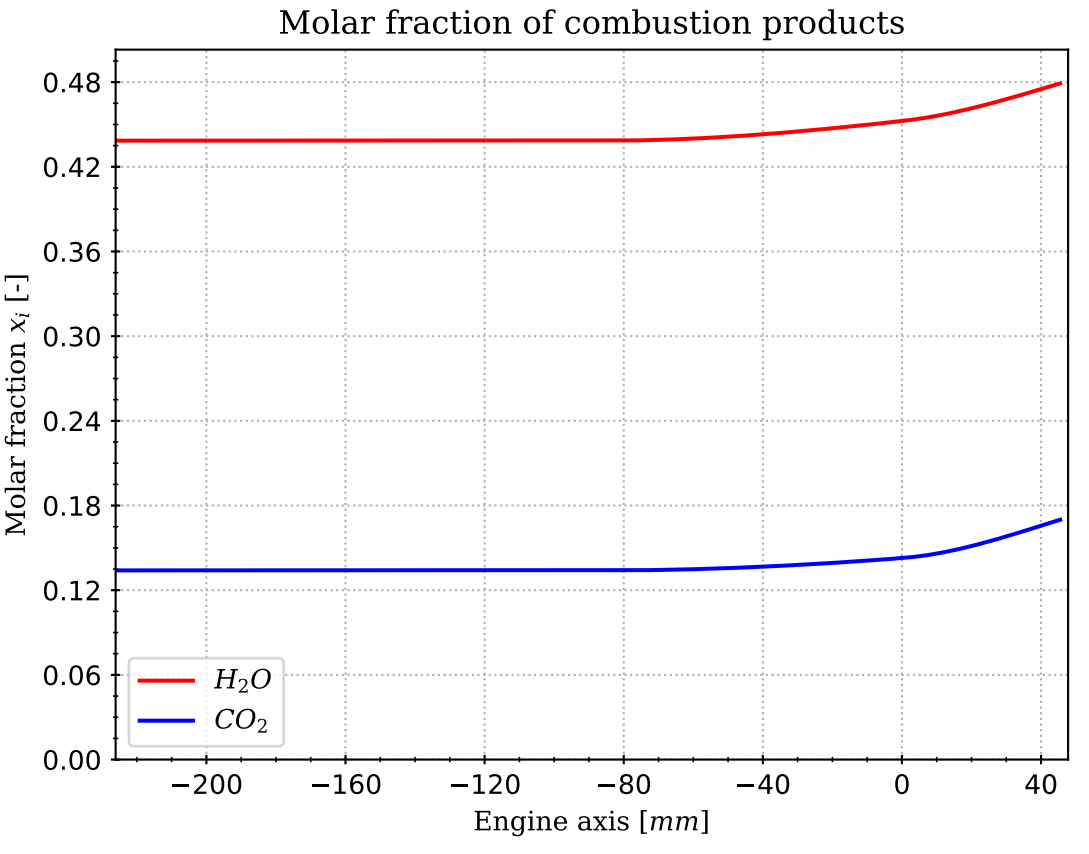

Hot gas conductivity
0.345
0.330
0.315
0.300
0.285
0.270
0.255
0.240
0.225
λ $[\frac{W}{m \cdot K}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

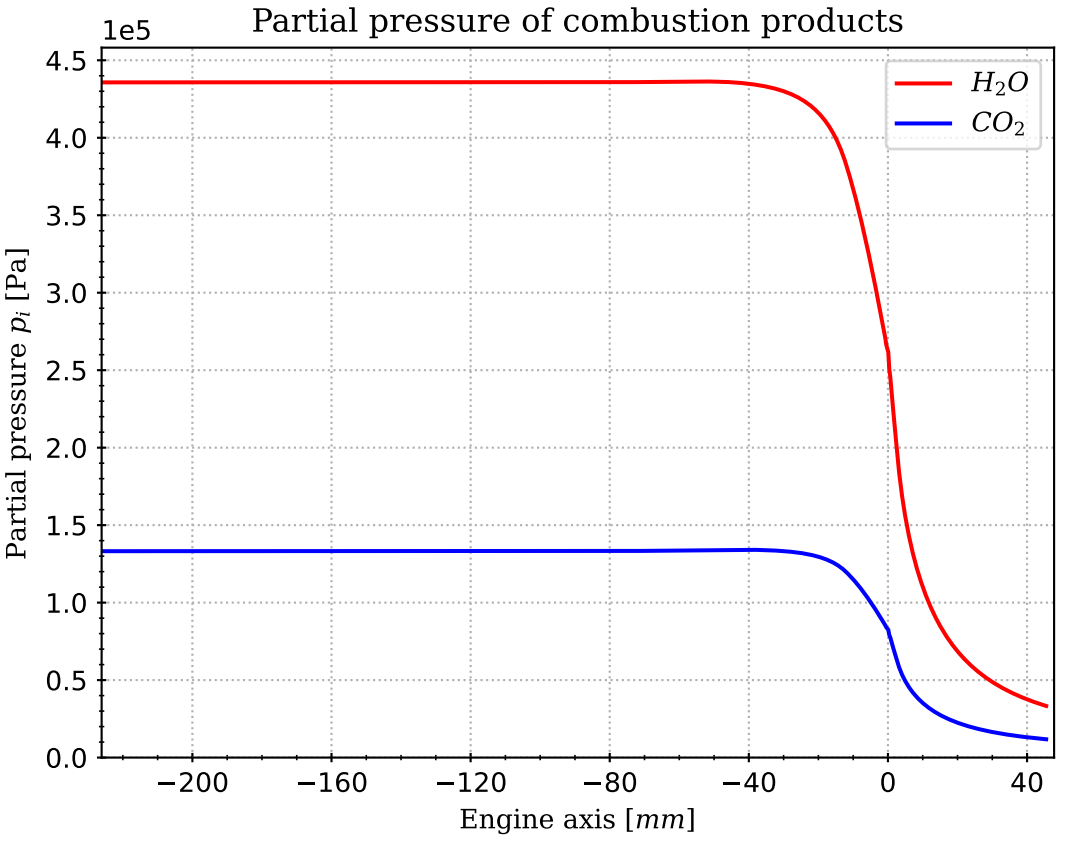

Hot gas Prandtl number
0.71
0.70
0.69
0.68
0.67
0.66
0.65
0.64
0.63
Pr_g [-]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

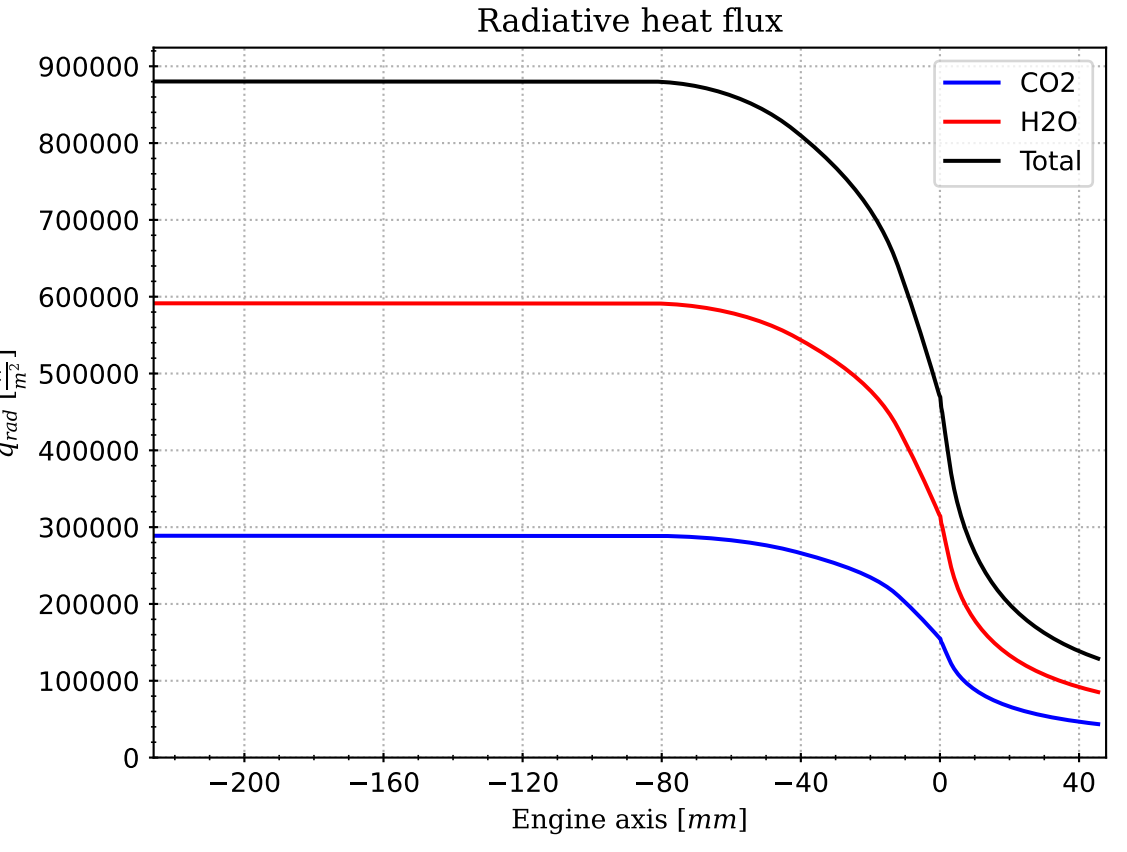

Hot-side convection coefficient h_g
5500
5000
4500
4000
3500
3000
2500
2000
1500
P [Pa]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

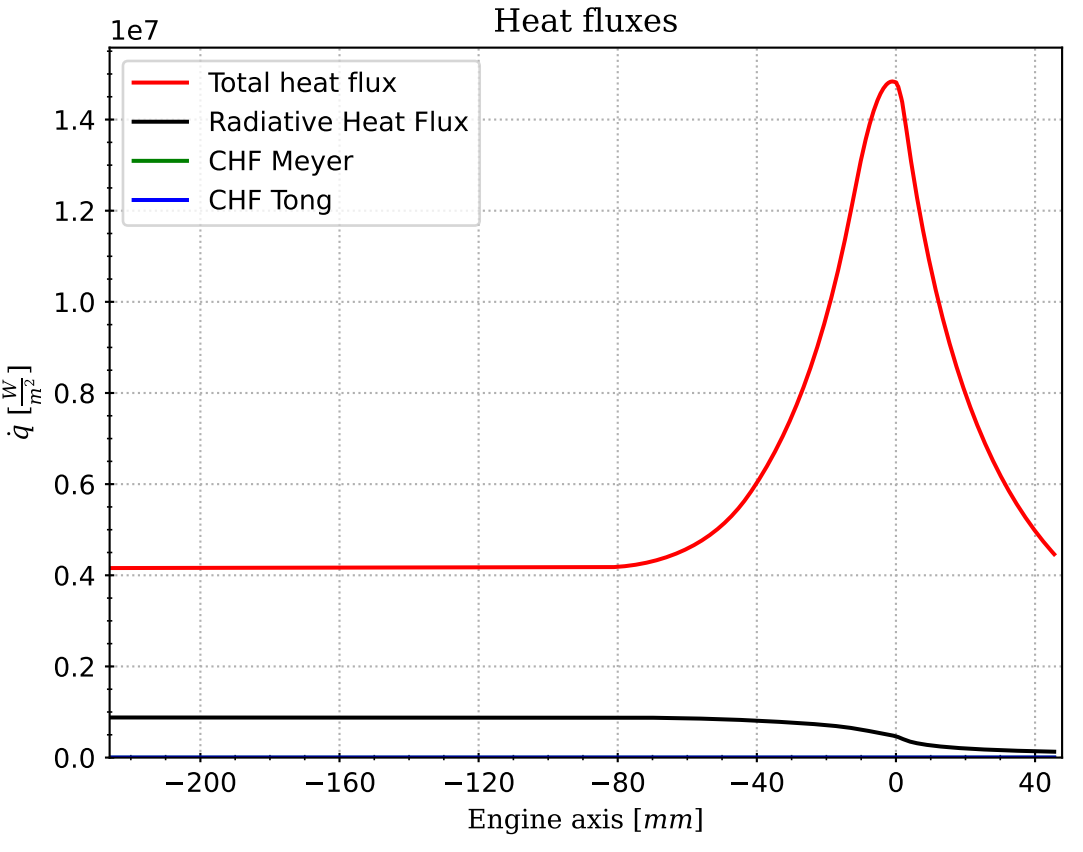

Bartz equation coefficient σ
1.550
1.525
1.500
1.475
1.450
1.425
1.400
1.375
1.350
1.325
σ [-]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

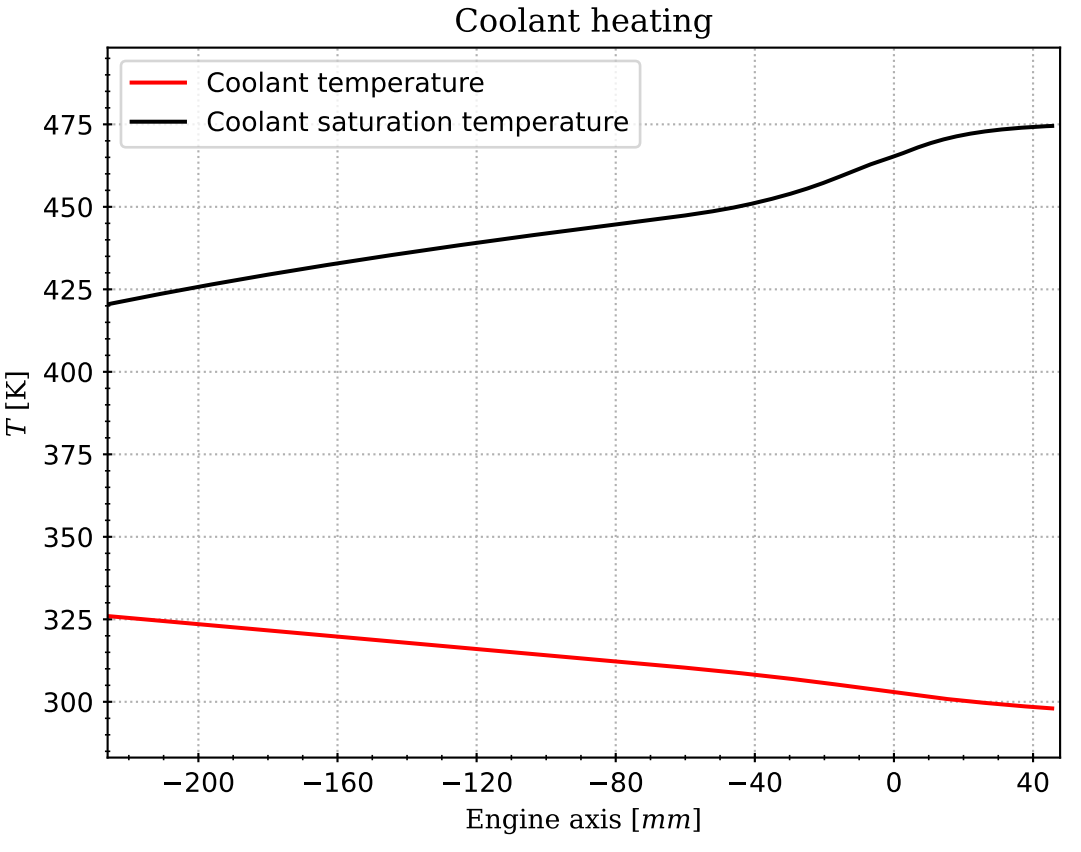

Cold-side convective coefficient h_l/h_{tp}
Gnielinski
Fin effect correction (Popp & Schmidt)
Subcooled flow boiling (Lui & Winterton)
270000
240000
210000
180000
150000
120000
90000
60000
30000
0
$h\ [\frac{W}{m^2 \cdot K}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

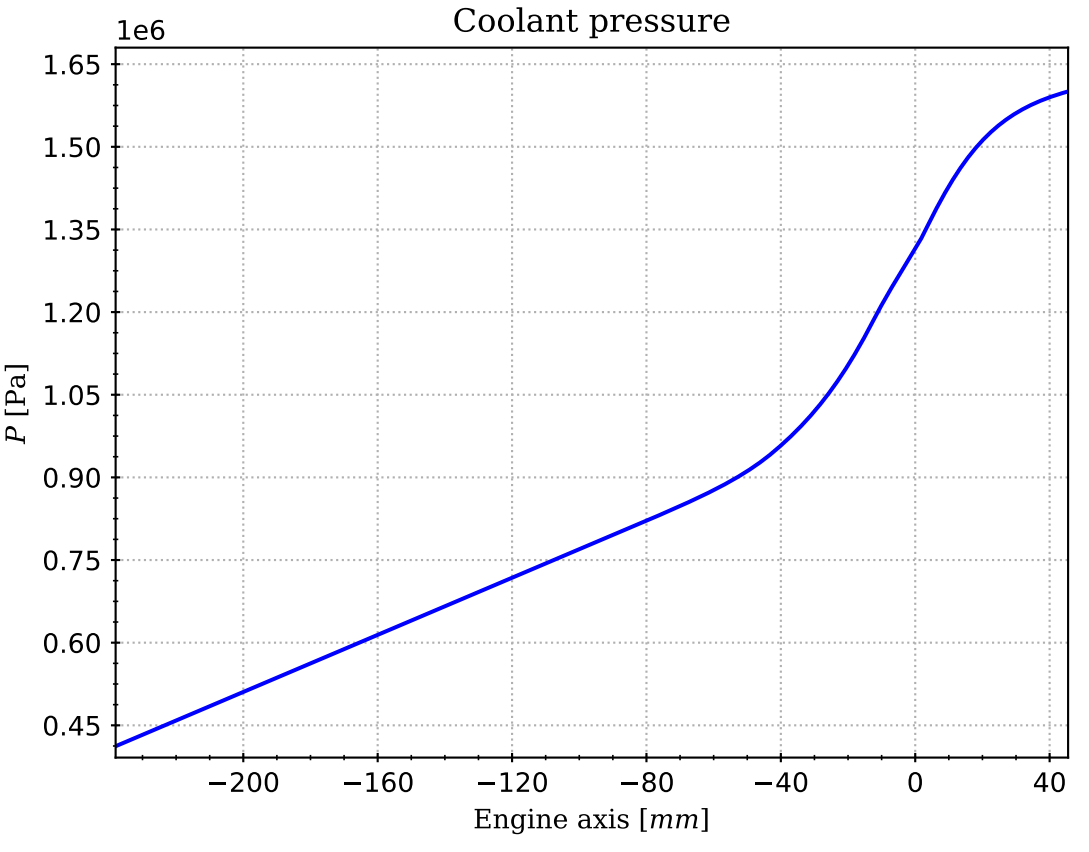

Molar fraction of combustion products
Molar fraction x_i [-]
0.48
0.42
0.36
0.30
0.24
0.18
0.12
0.06
0.00
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
H_2O
CO_2

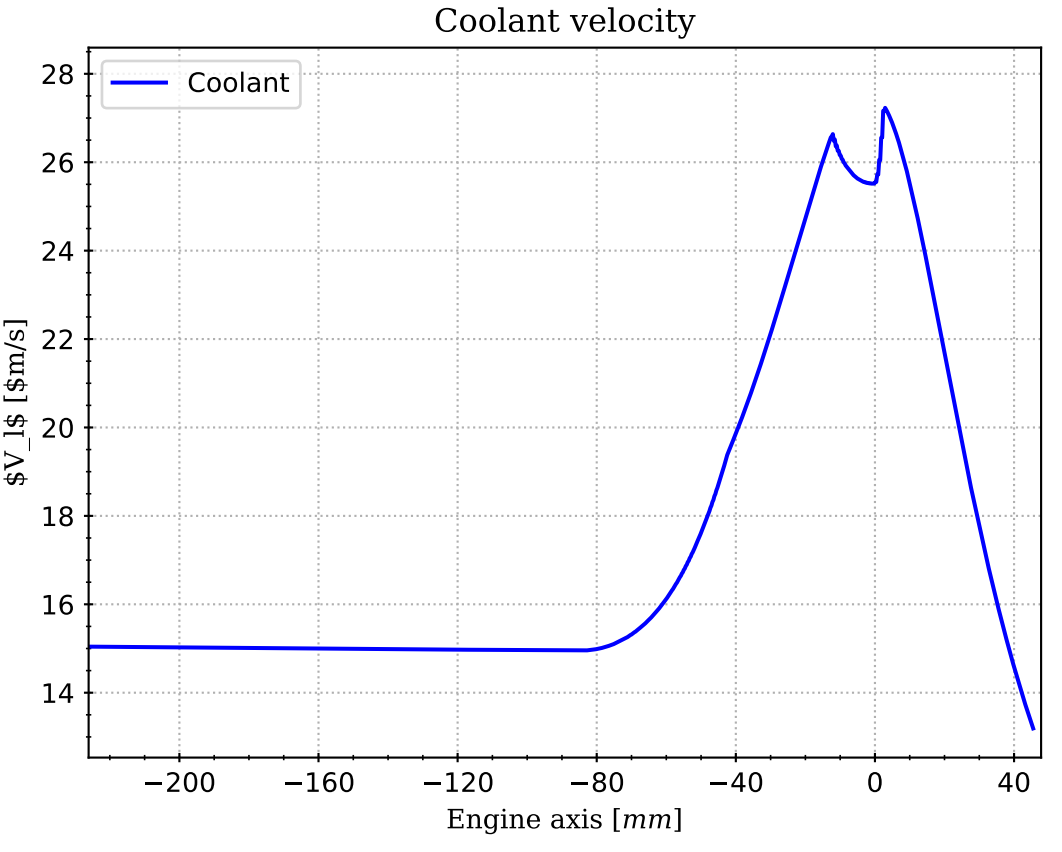

Partial pressure of combustion products
1e5
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Partial pressure p_i [Pa]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]
H_2O
CO_2

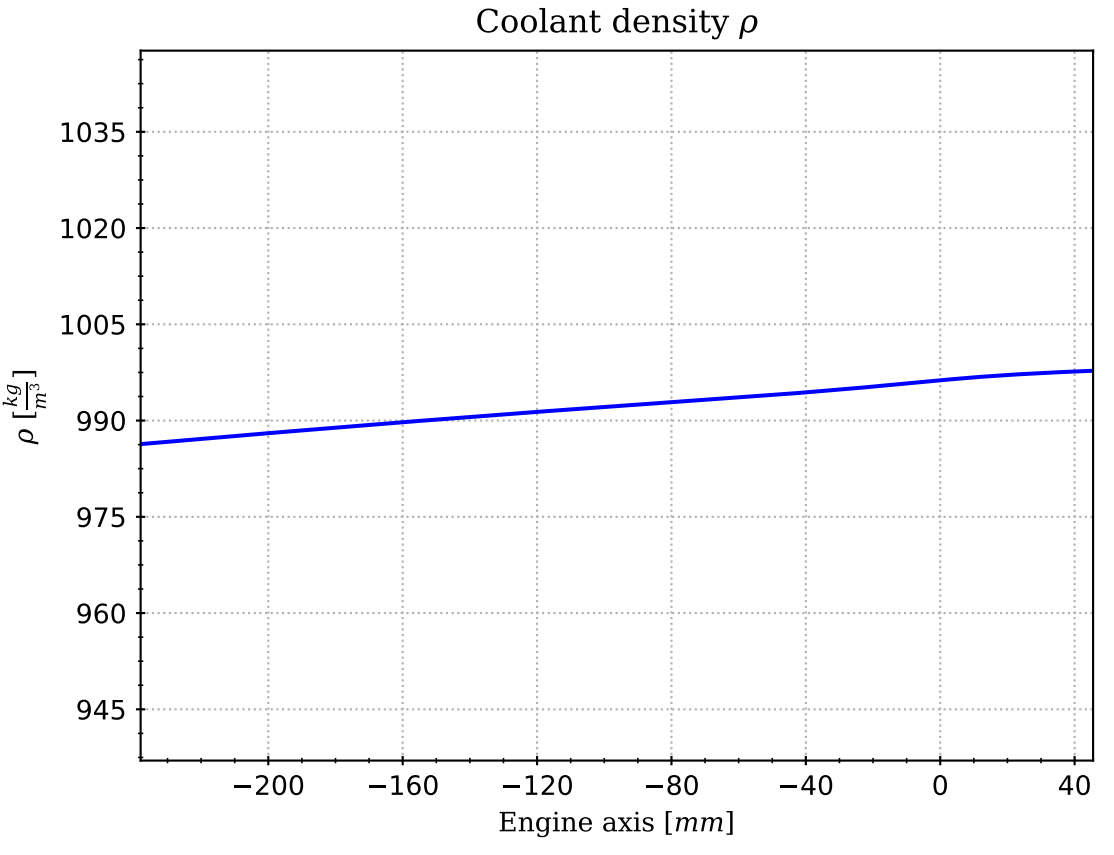

Radiative heat flux
CO2
H2O
Total
900000
800000
700000
600000
500000
400000
300000
200000
100000
0
q_{rad} $[\frac{W}{m^2}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

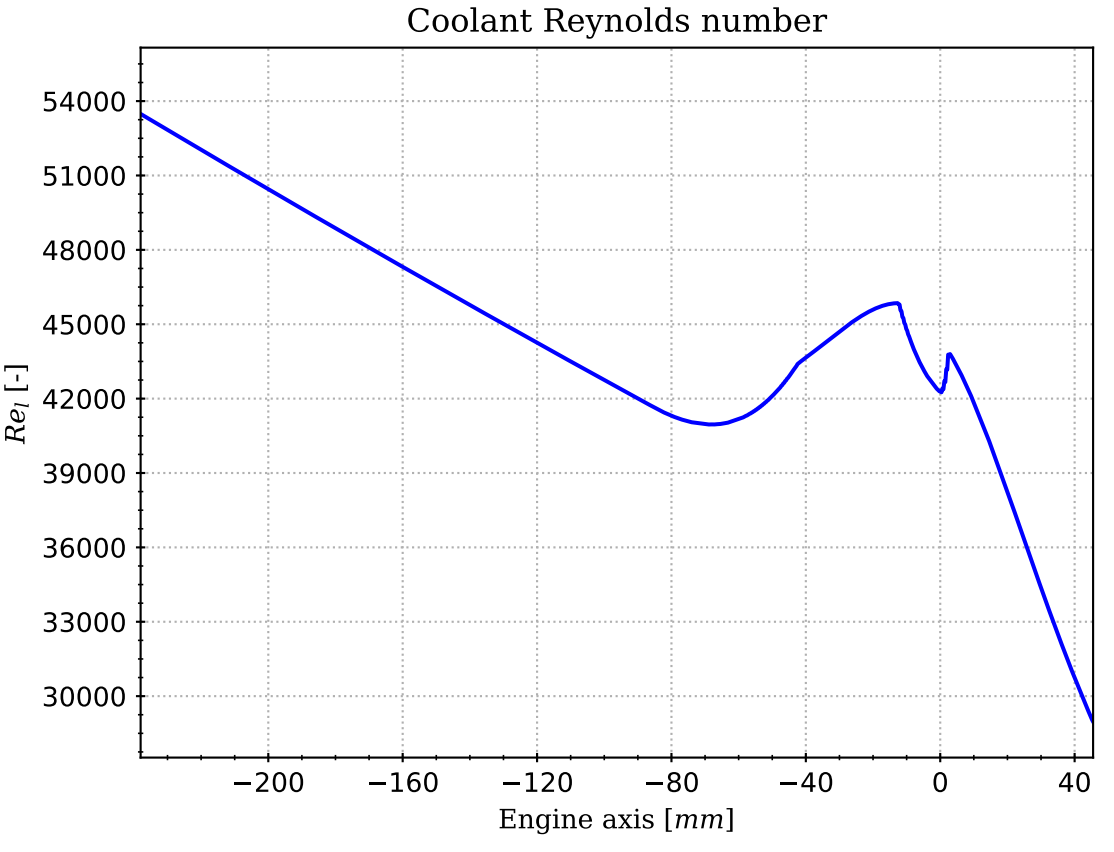

Heat fluxes
1e7
Total heat flux
Radiative Heat Flux
CHF Meyer
CHF Tong
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$\dot{q}\ [\frac{W}{m^2}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

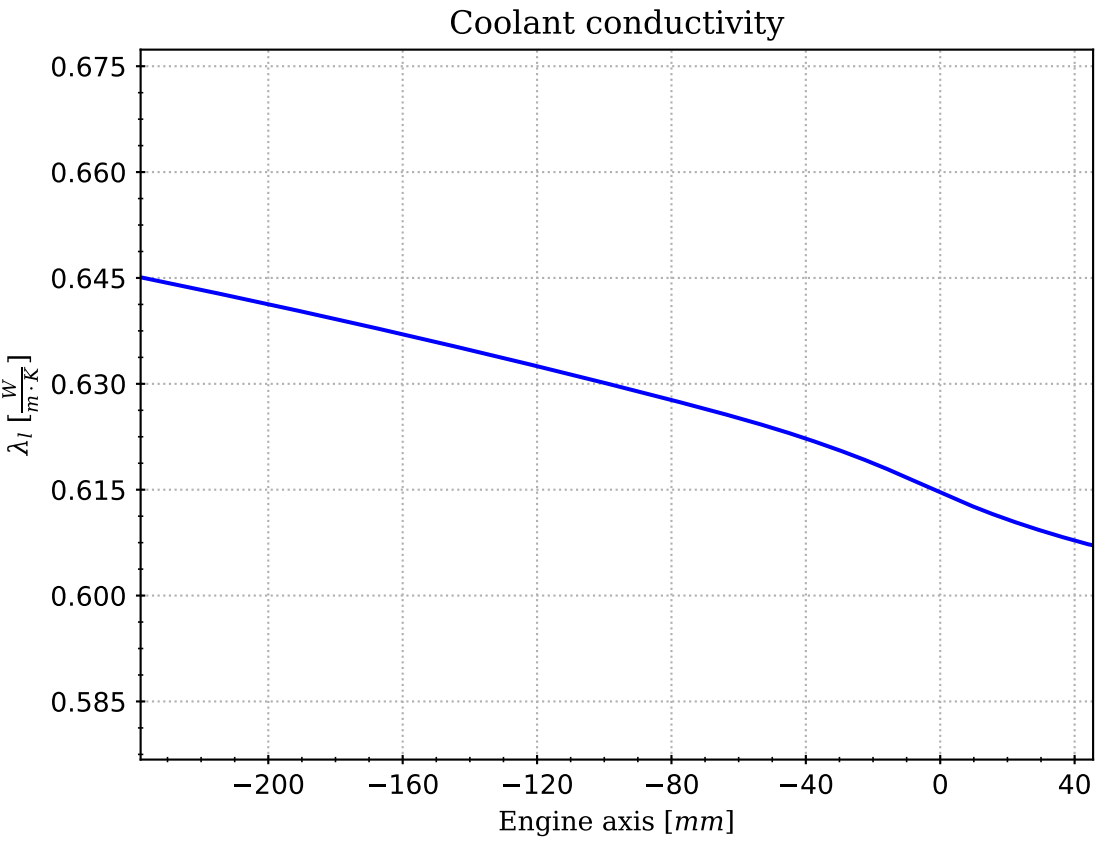

Coolant heating
Coolant temperature
Coolant saturation temperature
475
450
425
400
375
350
325
300
T [K]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

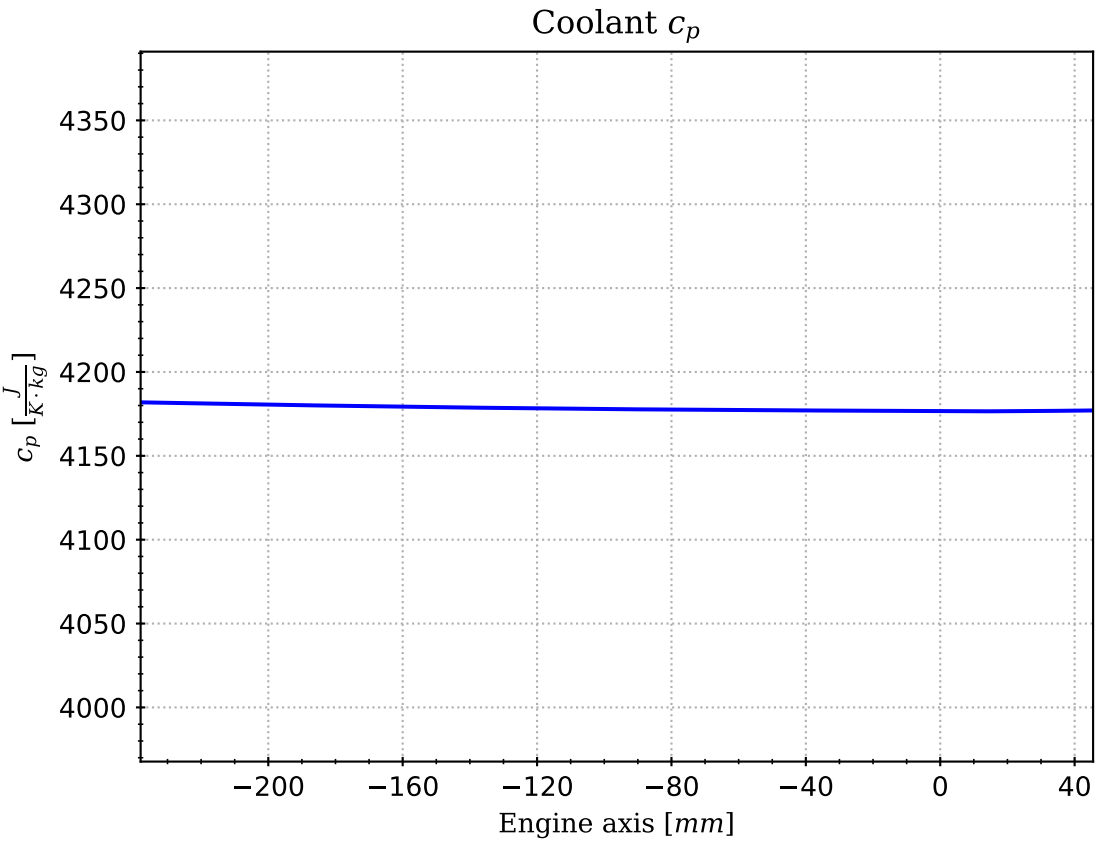

Coolant pressure
1e6
1.65
1.50
1.35
1.20
1.05
0.90
0.75
0.60
0.45
P [Pa]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

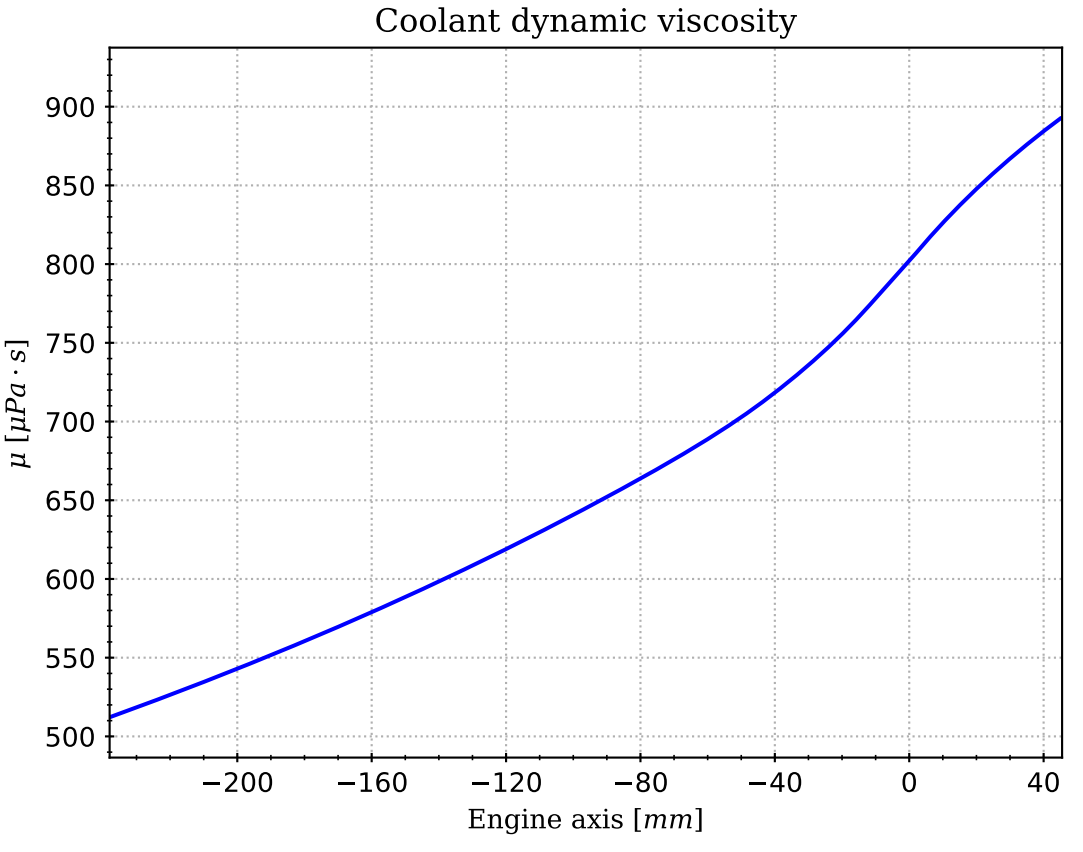

Coolant velocity
Coolant
28
26
24
22
20
18
16
14
V_l [$m/s]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

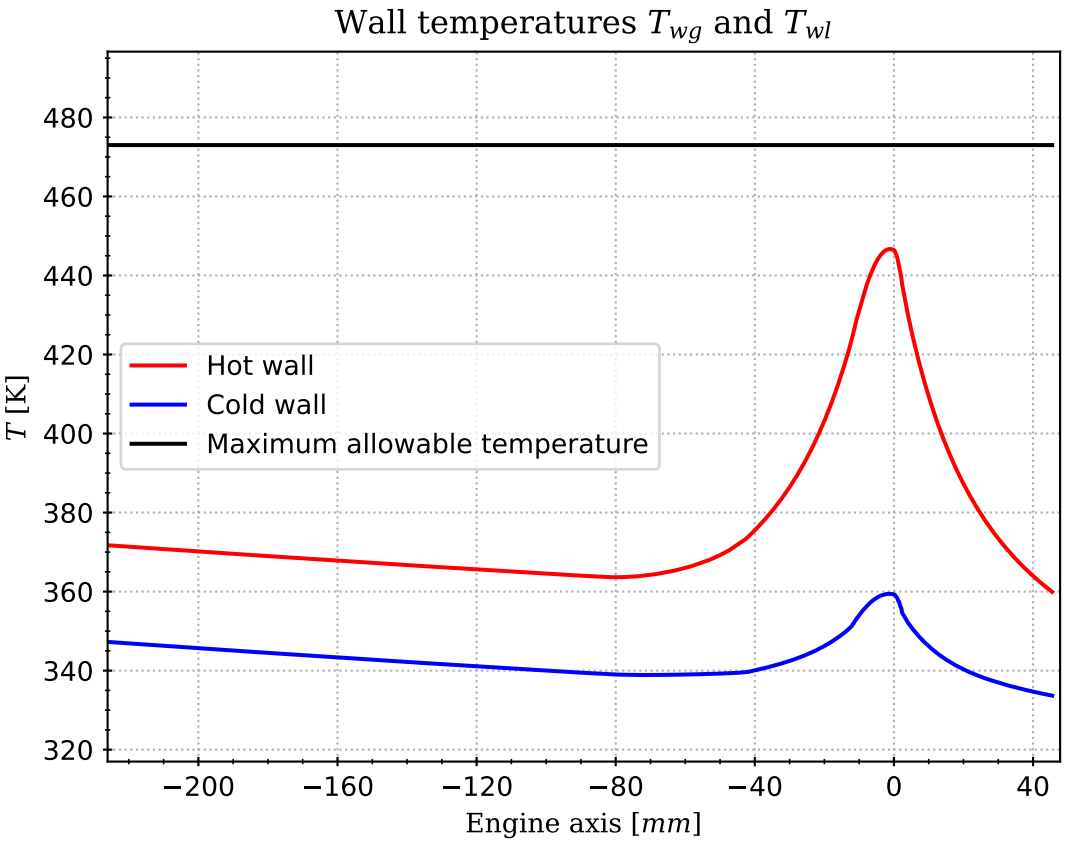

Coolant density ρ
1035
1020
1005
990
975
960
945
$\rho\ [\frac{kg}{m^3}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$

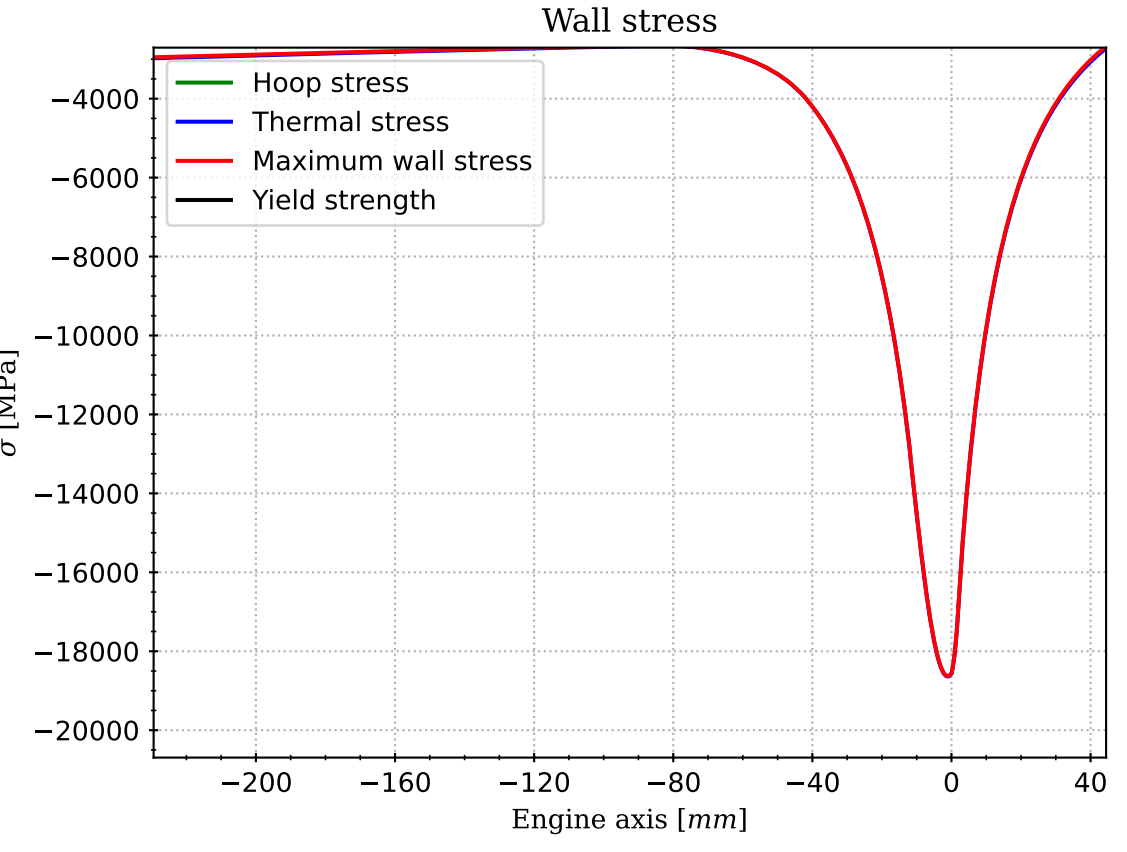

Coolant Reynolds number
54000
51000
48000
45000
42000
39000
36000
33000
30000
Re_l [-]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]

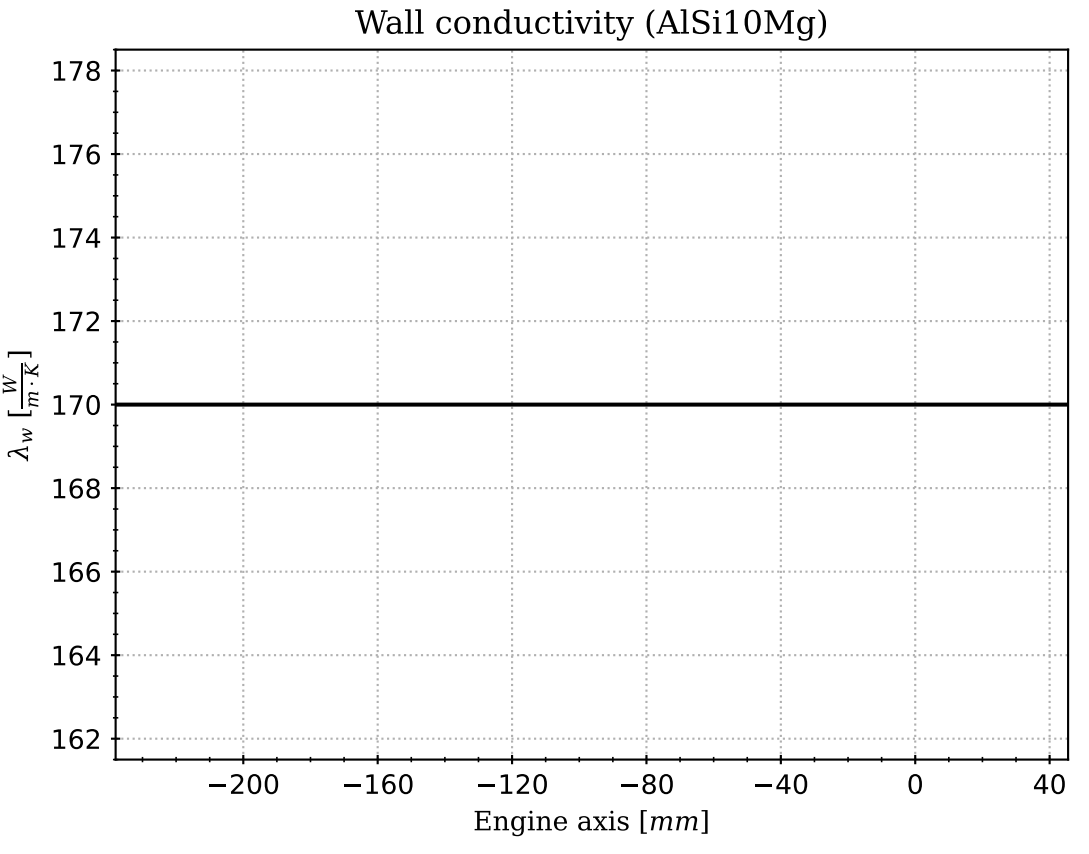

Coolant conductivity
0.675
0.660
0.645
0.630
0.615
0.600
0.585
λ_l $[\frac{W}{m \cdot K}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$


Coolant c_p
4350
4300
4250
4200
4150
4100
4050
4000
c_p $[\frac{J}{K \cdot kg}]$
−200
−160
−120
−80
−40
0
40
Engine axis $[mm]$


Coolant dynamic viscosity
900
850
800
750
700
650
600
550
500
μ [$\mu Pa \cdot s$]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]


Wall temperatures T_{wg} and T_{wl}
Hot wall
Cold wall
Maximum allowable temperature
T [K]
480
460
440
420
400
380
360
340
320
−200
−160
−120
−80
−40
0
40
Engine axis [mm]


Wall stress
Hoop stress
Thermal stress
Maximum wall stress
Yield strength
−4000
−6000
−8000
−10000
−12000
−14000
−16000
−18000
−20000
σ [MPa]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]


Wall conductivity (AlSi10Mg)
178
176
174
172
170
168
166
164
162
λ_w [$\frac{W}{m \cdot K}$]
−200
−160
−120
−80
−40
0
40
Engine axis [mm]